

LEPHALALE, South Africa

WMO: 681550

Lat: 23.6767S

Lon: 27.7050E

Elev: 2753

StdP: 13.29

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 40.1       | 42.5 | 8.8                           | 9.6 | 70.1 | 15.4 | 13.3 | 65.6 | 12.1                  | 66.0 | 10.6 | 66.8 | 2.0                   | 190  | 0.291 |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 2             | 19.2                      | 99.2            | 66.1 | 96.6 | 66.0 | 94.2 | 66.3 | 75.0                | 85.8 | 73.8 | 84.3 | 72.7 | 83.0 | 5.0                     | 0    |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 72.5                            | 133.6 | 78.8 | 71.1 | 127.3 | 78.1 | 69.9 | 121.7 | 77.3 | 40.6          | 85.4 | 39.5 | 84.5 | 38.3 | 83.1 | 79.5              |

Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |  |
| 11.3              | 9.7  | 8.6 | DB                         | 35.3 | 104.2              | 2.5 | 3.4                                                | 33.5 | 106.7      | 32.0 | 108.7      | 30.6 | 110.6      | 28.7 | 113.1 |  |
|                   |      |     | WB                         | 29.6 | 75.7               | 1.6 | 2.1                                                | 28.5 | 77.2       | 27.6 | 78.4       | 26.7 | 79.6       | 25.6 | 81.1  |  |

Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 72.2        | 79.4  | 79.9  | 77.0  | 71.4  | 65.9  | 60.5  | 59.9  | 65.4  | 72.7  | 76.9  | 78.4  | 79.2  |
|                                                                        | DBStd       | 8.57        | 4.13  | 4.11  | 4.37  | 4.31  | 4.61  | 3.56  | 3.42  | 5.04  | 5.33  | 6.05  | 5.61  | 5.19  |
|                                                                        | HDD50       | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 412         | 0     | 0     | 1     | 3     | 43    | 140   | 160   | 56    | 6     | 2     | 1     | 0     |
|                                                                        | CDD50       | 8095        | 912   | 836   | 837   | 642   | 494   | 316   | 308   | 478   | 682   | 832   | 853   | 905   |
|                                                                        | CDD65       | 3030        | 447   | 416   | 372   | 195   | 71    | 6     | 3     | 70    | 237   | 370   | 403   | 440   |
|                                                                        | CDH74       | 30371       | 4248  | 3981  | 3146  | 1453  | 781   | 243   | 222   | 988   | 2601  | 4070  | 4226  | 4412  |
|                                                                        | CDH80       | 13381       | 1847  | 1796  | 1330  | 492   | 184   | 25    | 18    | 312   | 1222  | 2034  | 2063  | 2058  |
| (14) Wind                                                              | WSAvg       | 3.5         | 3.5   | 3.4   | 3.1   | 2.9   | 2.6   | 2.7   | 3.1   | 3.5   | 4.4   | 4.7   | 4.3   | 3.7   |
| (15) Precipitation                                                     | PrecAvg     | 15.9        | 3.0   | 2.3   | 2.0   | 1.1   | 0.2   | 0.2   | 0.0   | 0.1   | 0.2   | 1.3   | 2.3   | 3.3   |
|                                                                        | PrecMax     | 23.8        | 5.7   | 4.3   | 7.7   | 3.3   | 1.5   | 2.6   | 0.5   | 0.5   | 1.1   | 3.3   | 6.5   | 6.6   |
|                                                                        | PrecMin     | 8.7         | 0.7   | 0.7   | 0.2   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 1.3   |
|                                                                        | PrecStd     | 4.0         | 1.6   | 1.2   | 1.9   | 1.0   | 0.4   | 0.6   | 0.1   | 0.1   | 0.2   | 0.9   | 1.5   | 1.4   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 100.3 | 100.2 | 97.2  | 93.7  | 89.4  | 82.8  | 82.4  | 91.1  | 97.8  | 102.7 | 102.4 | 101.1 |
|                                                                        |             | MCWB        | 67.8  | 68.7  | 65.8  | 65.4  | 60.7  | 56.2  | 53.9  | 58.5  | 61.2  | 62.6  | 64.4  | 69.6  |
|                                                                        | 2%          | DB          | 95.1  | 95.8  | 93.6  | 89.5  | 84.5  | 79.4  | 79.1  | 87.2  | 94.3  | 98.0  | 97.6  | 96.7  |
|                                                                        |             | MCWB        | 69.4  | 68.8  | 66.5  | 64.9  | 60.1  | 55.8  | 54.2  | 57.9  | 60.2  | 62.7  | 65.6  | 68.9  |
|                                                                        | 5%          | DB          | 92.0  | 92.9  | 90.9  | 85.8  | 81.4  | 76.8  | 76.5  | 83.7  | 91.3  | 94.4  | 94.0  | 93.5  |
|                                                                        |             | MCWB        | 69.2  | 69.2  | 66.5  | 64.8  | 59.3  | 55.3  | 54.8  | 56.9  | 59.3  | 63.2  | 65.8  | 69.3  |
|                                                                        | 10%         | DB          | 89.3  | 89.8  | 87.5  | 82.3  | 78.5  | 73.6  | 73.5  | 79.6  | 87.7  | 90.7  | 90.8  | 90.3  |
|                                                                        |             | MCWB        | 69.3  | 68.4  | 67.0  | 64.6  | 58.9  | 55.0  | 54.2  | 56.2  | 59.0  | 62.8  | 65.9  | 69.0  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 77.4  | 75.4  | 74.8  | 74.9  | 70.2  | 61.7  | 60.1  | 64.6  | 66.3  | 69.3  | 74.1  | 76.3  |
|                                                                        |             | MCDB        | 85.6  | 88.3  | 82.8  | 79.7  | 78.8  | 74.3  | 72.6  | 78.3  | 86.3  | 86.3  | 88.1  | 89.0  |
|                                                                        | 2%          | WB          | 74.8  | 74.0  | 72.9  | 71.9  | 66.3  | 59.4  | 58.4  | 61.8  | 64.3  | 67.9  | 71.9  | 74.7  |
|                                                                        |             | MCDB        | 85.2  | 85.7  | 80.8  | 78.8  | 72.7  | 72.7  | 71.7  | 78.6  | 84.7  | 84.6  | 83.8  | 86.0  |
|                                                                        | 5%          | WB          | 73.4  | 72.8  | 71.6  | 69.2  | 62.7  | 58.1  | 57.0  | 60.0  | 62.9  | 66.6  | 70.2  | 73.1  |
|                                                                        |             | MCDB        | 85.1  | 83.4  | 80.7  | 78.4  | 75.7  | 71.0  | 70.8  | 77.4  | 82.3  | 83.2  | 82.6  | 83.6  |
|                                                                        | 10%         | WB          | 72.0  | 71.6  | 70.2  | 67.0  | 61.0  | 56.6  | 55.6  | 58.3  | 61.5  | 65.5  | 69.0  | 71.7  |
|                                                                        |             | MCDB        | 82.6  | 81.5  | 78.5  | 77.3  | 74.4  | 69.3  | 69.5  | 75.7  | 80.2  | 82.2  | 82.0  | 82.6  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 18.4  | 19.2  | 19.6  | 21.9  | 27.2  | 29.1  | 29.0  | 29.2  | 27.8  | 23.9  | 21.5  | 19.2  |
|                                                                        |             | MCDBR       | 23.7  | 24.2  | 25.7  | 27.8  | 30.6  | 33.5  | 33.7  | 33.9  | 33.3  | 29.7  | 26.6  | 24.5  |
|                                                                        | 5% WB       | MCWBR       | 6.2   | 6.5   | 6.9   | 8.8   | 11.1  | 14.6  | 15.5  | 13.5  | 11.9  | 9.0   | 6.9   | 6.1   |
|                                                                        |             | MCWBR       | 18.8  | 18.1  | 17.5  | 19.8  | 23.5  | 26.2  | 27.1  | 27.6  | 26.6  | 23.1  | 21.4  | 19.7  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.394 | 0.396 | 0.385 | 0.355 | 0.314 | 0.300 | 0.296 | 0.354 | 0.409 | 0.429 | 0.394 | 0.394 |
|                                                                        |             | taud        | 2.376 | 2.369 | 2.382 | 2.430 | 2.503 | 2.513 | 2.516 | 2.341 | 2.207 | 2.201 | 2.344 | 2.376 |
|                                                                        | Ebn at Noon | Ebn at Noon | 301   | 296   | 291   | 286   | 286   | 285   | 290   | 282   | 279   | 283   | 299   | 302   |
|                                                                        |             | Edh at Noon | 42    | 41    | 39    | 34    | 29    | 27    | 28    | 36    | 45    | 48    | 43    | 42    |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 2140  | 2015  | 1816  | 1563  | 1459  | 1301  | 1401  | 1658  | 1918  | 2073  | 2116  | 2102  |
|                                                                        | RadStd      | RadStd      | 163   | 200   | 181   | 138   | 59    | 77    | 37    | 75    | 100   | 128   | 196   | 139   |

Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (2 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page